Grade 6 Accelerated
Unit 14
Lesson 9 Practice

Name_Answer Key_________
Date______________________
Period_____________________

1. For each inequality, write three numbers that could represent the solution set. Answer may vary.

   a. $\frac{1}{2} < p$ $\frac{3}{4}, 1, 3$

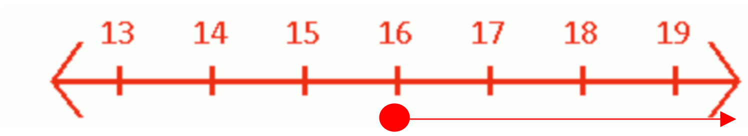
   b. $-3 \leq g$ $-2, 0, 4$

   c. $w < 10$ $9, 0, -1$

2. You have to be at least 18 years old to see the new scary movie at the movie theater. Write an inequality to represent the ages that one can be to see the scary movie in the theater. Let a represent the age.

   $a \geq 18$

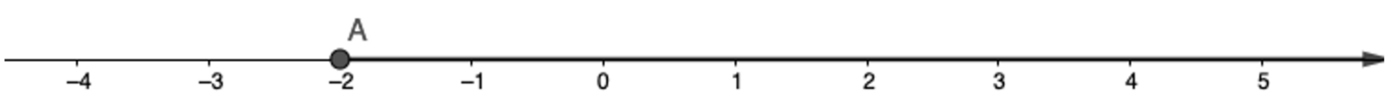
3. Graph the solution $16 \leq x$.

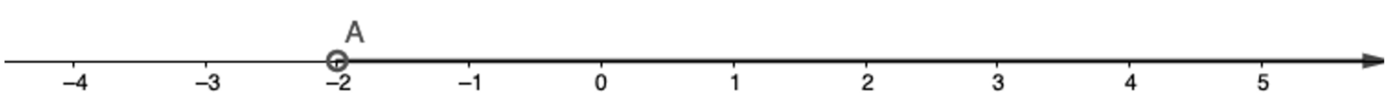


4. Stephanie states that the numbers $(6, 8, 10)$ are in the solution set for the inequality $6 \geq t$. Explain why Stephanie's answer is not accurate.

   6 is equal to t, but 8 and 10 are greater than 6.

5. Compare and contrast the two number lines.





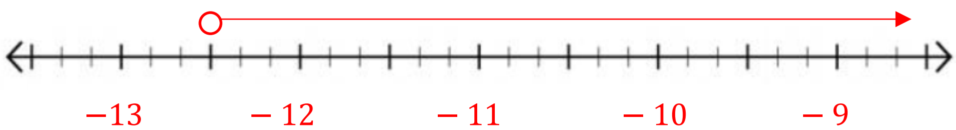
The first number line is greater than or equal to -2 and the second is greater than -2.

6. Use the inequality $-28.5 < v - 16$ to complete the following:

a. Find the solution set by isolating the variable. Show your work.

$$\begin{array}{l} -28.5 < v - 16 \\ \underline{\;+16 \qquad\quad +16} \\ -12.5 < v \text{ or } \quad v > -12.5 \end{array}$$

b. Graph the solution set on the number line.



c. Use substitution to test a value or values in order to verify that the solution and the graph are correct.

$v = -10$ $\qquad -28.5 < -10 - 16$

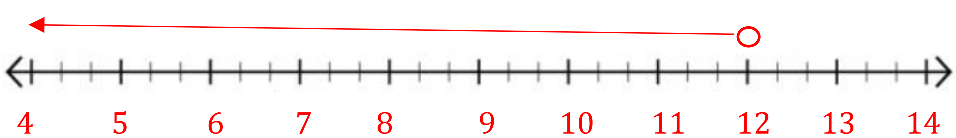
$\qquad\qquad -28.5 < -26$

7. How do you isolate a variable when solving an inequality?
Use the opposite operations just as you would when solving an equation.

Solve each inequality. Then, graph its solution set.

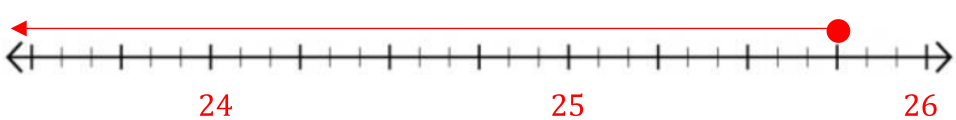
8. $6 > z - 6$

$$\begin{array}{l} \underline{+6 \qquad +6} \\ 12 > z \\ z < 12 \end{array}$$



9. $47\frac{3}{4} \geq w + 22$

$$\begin{array}{l} \underline{-22 \qquad\quad -22} \\ 25\frac{3}{4} \geq w \\ w \leq 25\frac{3}{4} \end{array}$$



10. $x + 8.2 < 18$

$$\begin{array}{l} \underline{-8.2 \quad -8.2} \\ x < 9.8 \end{array}$$